Aror University of Art, Architecture, Design & Heritage Sukkur

Lab # 05	**Application of ICT (Fall-2025)**
Course Title:	Application of ICT
Lab Instructor:	Rashida Naz
Credits (Theory + Lab):	3 Credit Hours (2 Hrs Theory and 1 Hr Lab)
Lab Duration:	2 Hrs
Lab Title:	Working with Excel

Objectives of the Lab:

In this lab, the students are going to learn:

1. To apply basic arithmetic formulas in Microsoft Excel
2. To learn about the basic formatting, cell splitting and merging along with basic data calculations

Basic Operations:

Nearly all the software in Microsoft Office package have the same tools for font color, background color and font size etc. with font background color replaced by cell color. We can perform these operations on cells and the text/ numbers within using the following tool bar:

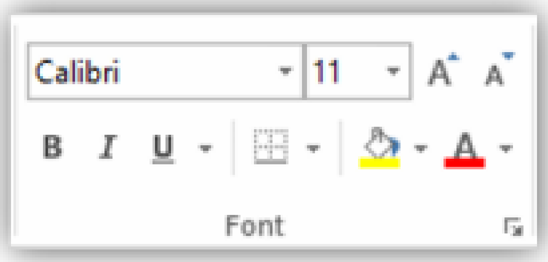

Figure 2: Font and Background Tools

Cell Merging and Splitting: We can also merge multiple cells if data takes larger space. Conversely, we can also unmerge them. This can be done using the merge tool in Alignment tab as shown in the figure:

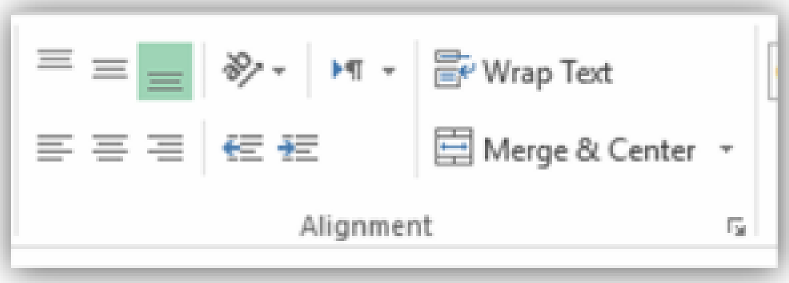

Figure 3: Cell Merge and Split Tool

Performing Arithmetic Operations: As we know that Microsoft Excel is ideal for office work therefore, it is well equipped for number crunching and calculation tools. We can perform basic arithmetic operations along with several useful data operations and the best thing is that we can do this to all the data at once. These operations include sum, average, minimum and maximum etc. For instance, we have two numbers in two separate cells. If we want to calculate and show their sum in the result cell, one way of doing it would be to write "=SUM (Cell 1: Cell 2)" in the result box. This is how formulae are entered in MS Excel i.e. using the following pattern:

=Formula name (Starting Cell number : Ending Cell Number)

This is demonstrated in the figure below:

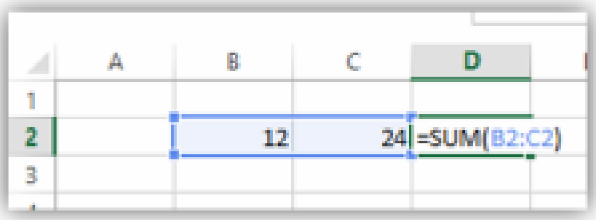

Figure 4: Entering Formula for Summation

As a side note, this process can extended for N number of cells (in a row or in a column). **Task 1: Formula Practice**

Download the excel file titled as "Excel Formulae Practice.xlsx". The spreadsheet is divided into four sections i.e. Addition, Mean, Sorting and min/max etc. Perform the appropriate action and then save it as "task1_lab3_cs".

Conditional Formatting:

Microsoft Excel has a cool feature called "Conditional Formatting". Using this tool, you can highlight data based on some condition. We can use these features from the Conditional Formatting bar as shown:

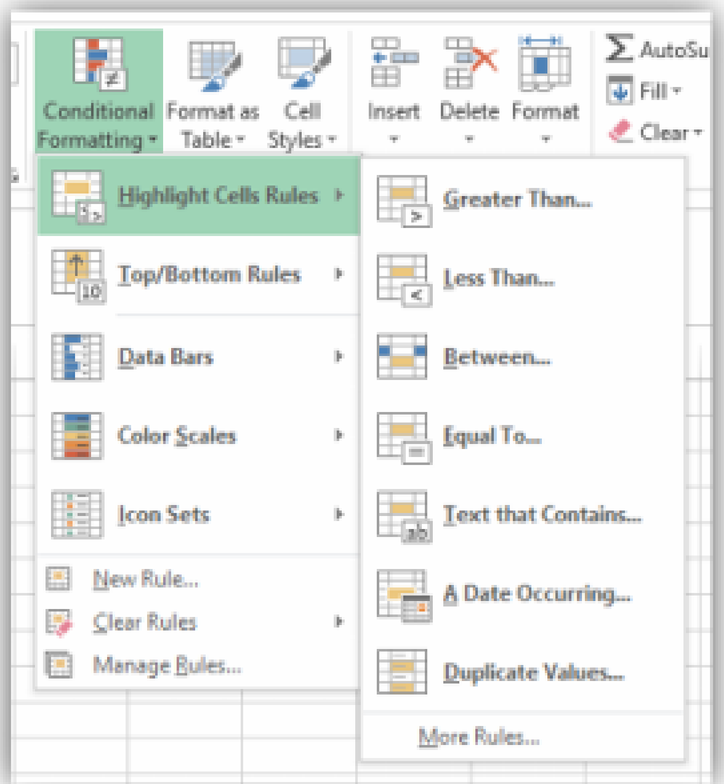

Figure 5: Conditional Formatting Tool

As you can see, there are multiple options available e.g. highlight cells based on some condition, we can also set data bars based on the numbers in a row or column and we can also set color scales based on numerical values etc. Let's perform another task which will include a basic use of this feature:

Task 2: Player Statistics

Download the file titled as "Cricket Players Excel Challenge.xlsx". In this file, the statistics of some notable players in the world is given. Perform the following operations on this dataset: 1. Compute the average score scored by each player. 2. Highlight the players younger than 30 years old using conditional formatting. 3. Find the minimum and maximum score in each category.

Charts in Microsoft Excel: Charts and diagrams are a great way of visualizing data and finding trends. In this part, we'll learn how to create chart, to stylize it in different ways, add legend entries, to alter the x and y axis data sets and to add trend lines etc. Let's get to it then: **Creating charts in Excel:** To create a chart in Microsoft Excel, we have to select two columns of data because there are two axes on a chart in case of bar, histogram or column type charts. We can add columns in case of other types such as pie charts etc. After selecting the data set, we have to go to the Insert menu and select chart type. For this exercise, we will choose a column chart. Taking the spreadsheet used in the task given above, we must first select two columns. To insert a column chart for this data, we go to the Insert menu, move to charts tab and select the column chart. The following graph should appear on your screen:

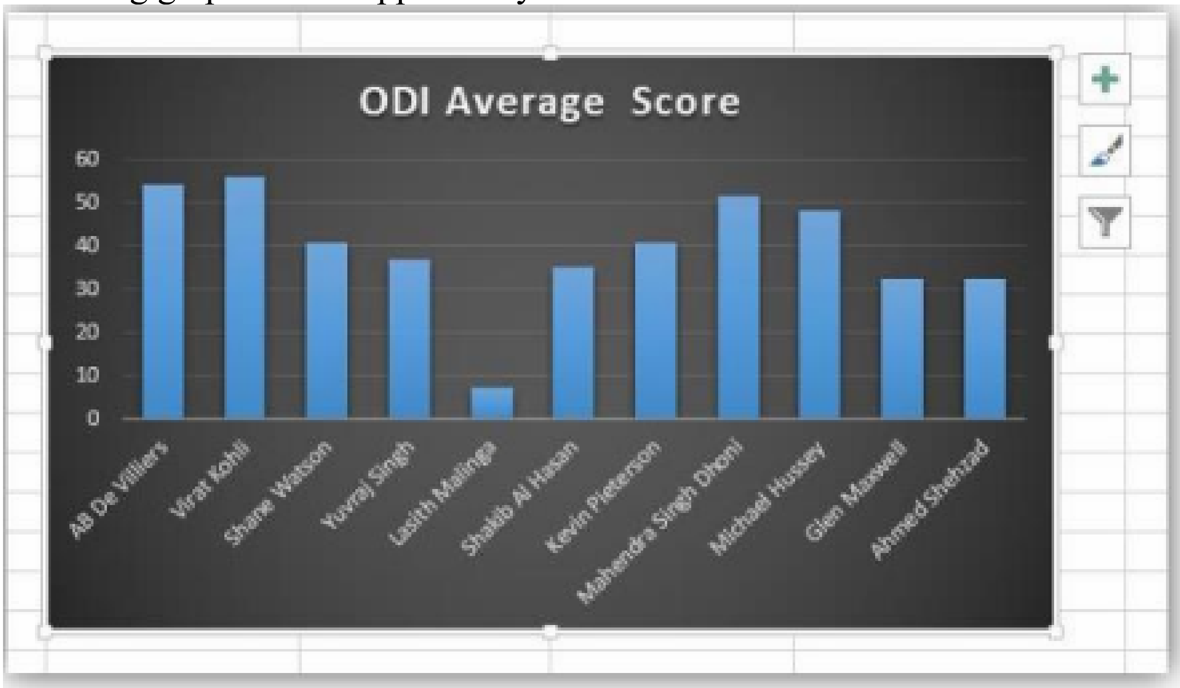

Figure 6: Simple Column Chart

Setting the name of chart and axes of the chart: To set the name of the chart, double click on the area where “ODI Average Score” is written and type in your desired name of the chart. If you look closely, you will notice a plus sign at the top right of the chart. If we click on it, a drop-down menu will appear. Check the box titled Axis Title and then add the names of the axes

drawn in the chart. In the figure given below, you can see the chart title and axes titles enclosed in a blue square:

Figure 7: Chart and Axis Title

Drawing comparative graphs and adding legends:

Sometimes, you have one data set on one axis and two data sets on the other and you want to visualize a comparative graph between the former one and the latter two. For this purpose, you’ll have to select the three columns and repeat the process for column plot. For example, we have ODI Score, and we want to see its comparison with Test Score. After repeating the procedure for column chart, a chart resembling the following figure will appear on screen:

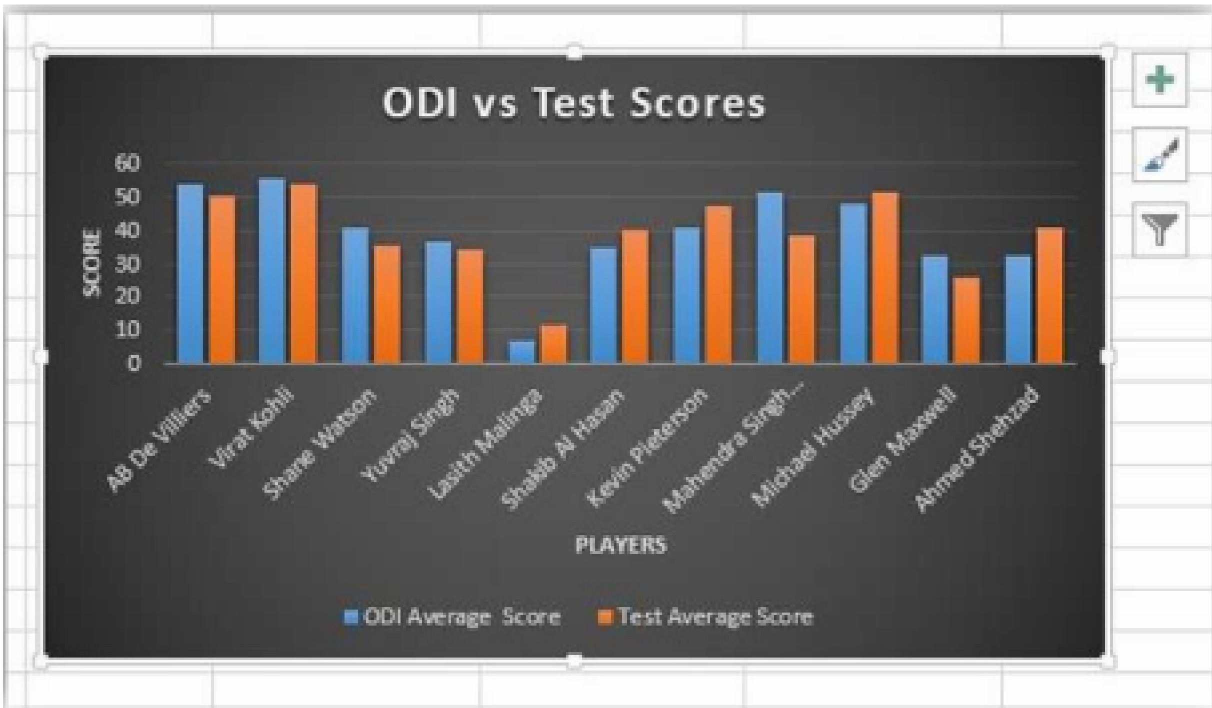

Figure 8: Comparative Chart Example

Task 3: Comparative Column Charts

You are given an MS Excel spreadsheet titled "Charts Practice Sheet.xlsx". The sheet contains list of countries that won gold, silver and bronze medals in Olympics 2004 in Athens. Using the column chart utility of Microsoft Excel, draw a comparative chart of three medals for the first ten countries. Give the chart title "Countries Performance Chart". Furthermore, add appropriate axis titles along with legend for elaboration. Save the file appropriately with your name and lab and task number.

Line Charts and Trend lines in Microsoft Excel: Previously, we looked at column charts and performed some exercises on it. In this part, we are going to use Line chart to visualize data and we'll use the trend line feature of Microsoft Excel to see the behavior of the data. Creating a Line Chart: To draw a line chart, you just have to select the data you want to plot, go to Insert menu and select line chart tab. This can be seen in the figure below: Adding Trend line and Equation: To add a trend line to our chart, we have to go to the top left of the screen and click on **"Add Chart Element"** dropdown menu. From there, we have select Trend Line and set the type to linear. We can also add the equation of the line by going to "More Trend Line Options" and checking **"Display Equation on Chart"** box.

Task 4: Line Charts and Trend Lines:

Download the spreadsheet "Line Chart Practice Sheet.xlsx". Plot a graph of the trainer's weight against the days. Add appropriate chart title and axis titles. Moreover, draw a trend line for the graph and also display its equation.